WEEK 6 PROJECT REPORT

Model Evaluation, Reporting & Business Impact Communication

Beauty & Wellness Market | Predictive Analytics

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Project / Model	Customer purchase-propensity prediction (illustrative use case)
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Prepared as a structured evaluation and business communication report.

Important note about the example results

This report is a ready-to-edit evaluation framework. No dataset,trainedmodel, or actual prediction results were provided when it was prepared. Therefore, all numerical scores, sample counts, and model comparisons shown below are clearly labelled **illustrative examples**, not measured project results. Replace them with values calculated from your own model before submitting the report as evidence of completed experiments. If you have not trained a model, retain the framework and label the results as a proposed evaluation plan.

Executive summary

Thisreport presents a practical framework for assessing a predictive machine-learning model for the beauty and wellness market. The example business problem is to estimate which customers are likely to purchase a product or service within a defined future period. Such a prediction can help a brand prioritise relevant offers, plan campaigns, improve customer engagement, and use marketing budgets more efficiently. The framework covers data preparation, validation, classification metrics, threshold selection, error analysis, model monitoring, and communication of results to business stakeholders.

The evaluation does not rely on accuracy alone. Precision, recall, F1-score, and AUC-ROC are considered because customer purchase data may be imbalanced and because false positive and false negative predictions have different business costs. The report also explains how model scores should be translated into campaign decisions, while protecting customer privacy and avoiding unfair or misleading targeting.

The sample performance table and chart use hypothetical values solely to demonstrate how results can be reported. A final business decision should be based on validated project results, campaign experiments, and measurable commercial outcomes such as incremental revenue, conversion lift, customer retention, and return on marketing spend.

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1. Project context and business problem

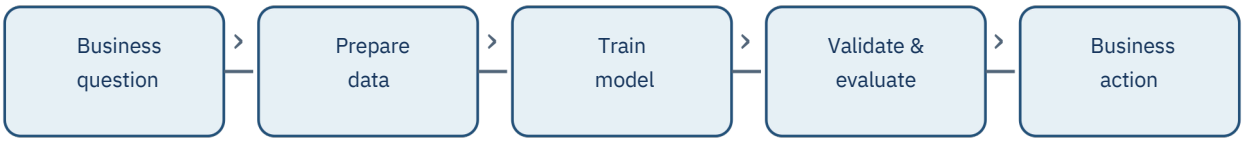
Beauty andwellness businessesoperate in a competitive market where customer needs, product preferences, purchase frequency, and campaign responses can vary. Brands may have customer interaction data from online stores, loyalty programmes, appointment bookings, product reviews, and marketing campaigns. When collected and used lawfully, these data can support better planning and more relevant customer communication.

Example prediction problem: predict whether an eligible customer will make a purchase within the next 30 days. The target is binary: 1 = purchase within 30 days; 0 = no purchase within 30 days. The prediction is intended to support campaign prioritisation, not to determine a customer's personal worth or access to essential services.

Business questions

- Which eligible customers are more likely to respond to a product or service campaign?
- Can marketing teams prioritise a limited campaign budget more effectively?
- Which customer groups or product categories show different response patterns?
- Does the model improve results compared with a simple baseline such as targeting all customers equally?

Evaluation workflow



The model should be judged against a simple baseline and against the actual business objective. A technically strong score is not sufficient if the model does not create incremental value, is too costly to operate, or produces unacceptable customer impact.

2. Evaluation objectives and success criteria

The evaluation framework hasfive objectives: (1) measurepredictive quality onunseen data; (2) assess the balance between missed buyers and incorrectly targeted non-buyers; (3) test whether performance is stable across different data samples and customer groups; (4) check whether predicted probabilities are useful for ranking and decision-making; and (5) connect model performance to business outcomes through a controlled campaign test.

Area	How it will be assessed	Example success condition
Predictive quality	Compare accuracy, precision, recall, F1 and AUC-ROC on a held-out test set.	Beats the baseline and meets pre-agreed minimums.
Generalisation	Use cross-validation and a time-based holdout where appropriate.	No large unexplained gap between validation and test scores.
Business usefulness	Run a controlled campaign and compare incremental conversion and revenue.	Positive incremental lift with acceptable cost.
Responsible use	Review data permissions, group-level errors, privacy and customer complaints.	No material unresolved privacy or fairness concern.

3. Data and modelling assumptions

A typicaldataset may contain customer-level records, purchase history, product or service category, campaign interactions, recency and frequency measures, and a clearly defined outcome window. Only data that the organisation is authorised to use should be included. Direct identifiers such as names, phone numbers, email addresses, and full addresses should be excluded from the modelling table unless strictly required for a lawful operational purpose; they

are generally not useful model features.

Suggested data preparation

- Define the prediction date and outcome window before creating features.
- Remove duplicate records and investigate missing or implausible values.
- Prevent data leakage: do not use information created after the prediction date.
- Encode categorical variables and scale numeric features where required by the chosen algorithm.
- Split records at customer level where repeated records could otherwise appear in both training and test sets.
- Document exclusions, transformations, feature definitions, and the dataset version.

A simple baseline should be created first. For example, the baseline may predict the majority class for every customer, or rank customers using a simple historical purchase-frequency rule. A machine-learning model should demonstrate a meaningful improvement over this baseline.

4. Evaluation metrics and their business meaning

Metric	What it measures	Business interpretation	Limitation
Accuracy	Share of all predictions that are correct.	Overall correctness across eligible customers.	Can look high when one outcome is much more common.
Precision	Among customers predicted to purchase, the share who actually purchase.	Higher precision can reduce wasted campaign contacts and offer costs.	May fall if the team tries to reach more potential buyers.
Recall	Among actual purchasers, the share correctly identified.	Higher recall means fewer likely buyers are missed.	May increase false positives and campaign cost.
F1-score	Harmonic mean of precision and recall.	Useful single summary when both missed buyers and unnecessary contacts matter.	Does not directly include campaign value or true negatives.
AUC-ROC	How well the model ranks positive cases above negative cases across thresholds.	Shows whether the model can separate likely purchasers from others.	Does not choose the operating threshold or show profit.
PR-AUC	Area under the precision-recall curve.	Often informative when purchases are rare.	Depends on the positive-class prevalence; interpret with context.
Calibration	Agreement between predicted probabilities and observed purchase rates.	A 0.70 score should correspond to roughly 70% outcomes in comparable groups.	Good ranking does not guarantee calibrated probabilities.

Metric selection rationale

Accuracy is reported for completeness, but it should not be the only decision metric. Precision matters when each contact, discount, sample, or promotion has a cost. Recall matters when missing a potential buyer is costly. F1-score provides a balance between precision and recall. AUC-ROC measures ranking ability across thresholds, while PR-AUC is especially useful if the positive purchase class is uncommon. Calibration is needed if teams interpret the model output as a probability.

For a campaign, the final threshold should be selected using expected business value, not automatically set to 0.50. For example, a brand may choose a higher threshold when discounts are expensive, or a lower threshold when the cost of reaching a customer is very small. The selected threshold and its expected cost-benefit trade-off must be documented.

5. Validation and testing framework

Arobust validation plan teststhe model instages. Thetest setmust remain untouched until model development and threshold decisions are complete. If the data are time-dependent, a chronological split is usually more realistic than a random split because it simulates predicting future customer behaviour from past information.

Stage	Procedure	Evidence to record
1. Data audit	Check target definition, missingness, duplicates, leakage, class balance and permitted data use.	Data dictionary, audit log, class distribution.
2. Baseline	Evaluate a majority-class or simple business-rule model.	Baseline confusion matrix and metrics.
3. Training	Train candidate algorithms using training data only; fit preprocessing within each training fold.	Model settings, feature list, training version.
4. Cross-validation	Use stratified k-fold for independent classification data, or time-series/forward chaining for temporal data.	Mean and spread of validation metrics.
5. Model selection	Compare candidates and tune hyperparameters without using the final test set.	Selection rationale and tuning record.
6. Final test	Evaluate the selected model once on a held-out test set representing unseen cases.	Final metrics, confusion matrix, ROC/PR curves.
7. Business pilot	Run a controlled campaign against a suitable control group.	Incremental conversion, revenue, margin, opt-outs and cost.
8. Monitoring	Track drift, data quality, performance and business outcomes after deployment.	Monitoring dashboard and retraining triggers.

Recommended split

As an initial design for a sufficiently large dataset, use approximately 70% of records for training, 15% for validation, and 15% for final testing. These percentages are not universal rules. Use a chronological split if future prediction is the real use case, and ensure that all records belonging to one customer stay within the same split when repeated customer records could cause leakage. If data are limited, use cross-validation within the development portion and preserve a final holdout.

Confusion matrix

	Predicted purchase	Predicted no purchase
Actual purchase	True positive (TP) Likely buyer correctly identified	False negative (FN) Likely buyer missed
Actual no purchase	False positive (FP) Unnecessary campaign contact	True negative (TN) Non-buyer correctly excluded

The confusion matrix makes errors understandable. False positives may waste campaign spend or annoy customers with irrelevant offers. False negatives may mean lost sales opportunities. The acceptable balance depends on the cost of a contact, the value of a purchase, discount margin, and customer experience.

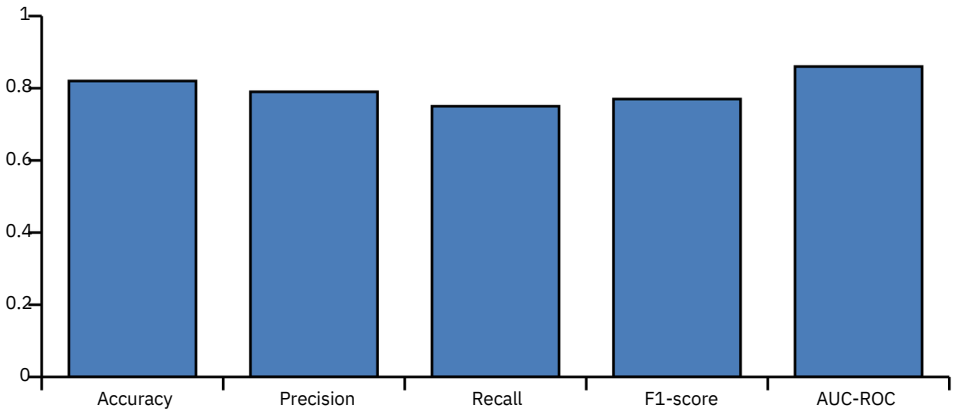
6. Illustrative results and visual reporting

Illustrative exampleonly: The valuesbeloware invented for demonstratingreport presentation. They are not actual results from a supplied dataset or trained model. Replace them with the model's measured test-set values.

Metric	Illustrative score	Plain-language meaning
Accuracy	0.82 (82%)	82 out of every 100 test predictions are correct overall.
Precision	0.79 (79%)	About 79 of 100 customers predicted to purchase actually purchase.
Recall	0.75 (75%)	The model identifies about 75 of every 100 actual purchasers.
F1-score	0.77	A combined measure of precision and recall.

Metric	Illustrative score	Plain-language meaning
AUC-ROC	0.86	The model shows useful ability to rank purchasers above non-purchasers.

Illustrative model evaluation scores



How to interpret this example

In this hypothetical example, the model has an AUC-ROC of 0.86, suggesting useful ranking ability. Precision of 79% suggests that a campaign based on positive predictions could reduce irrelevant contacts compared with a less selective approach. Recall of 75% means one quarter of actual purchasers are not identified at the selected threshold. Whether that is acceptable depends on the campaign's economics and the cost of missed opportunities. The business team should inspect results at multiple thresholds before choosing a campaign audience.

Recommended visualisations for the final project

- Confusion matrix heatmap: shows counts of true positives, false positives, true negatives and false negatives.
- ROC curve: plots true-positive rate against false-positive rate at different thresholds.
- Precision-recall curve: shows the trade-off between precision and recall, particularly useful for imbalanced outcomes.
- Calibration plot: compares predicted probability bands with observed purchase rates.
- Feature-importance chart: shows which input variables contribute most to model predictions; it should not be presented as proof of causation.
- Business impact chart: compares campaign conversion, incremental revenue, cost and margin for treatment and control groups.

7. Error analysis, fairness and risk controls

Asingle average score can hide important weaknesses. Review errors by relevant, lawful and sufficiently large customer segments, such as broad product categories, purchase-frequency bands, or new versus returning customers. Do not infer sensitive characteristics or use protected traits for targeting. Small groups should be handled carefully because estimates may be unstable.

Error review checklist

- Inspect false positives to learn whether promotions are being sent to customers who are unlikely to buy.
- Inspect false negatives to understand which potential purchasers the model misses.
- Compare precision, recall and calibration across relevant customer groups where lawful and appropriate.
- Check whether missing data or historical campaign practices create systematic disadvantages.
- Review customer complaints, opt-outs, unsubscribe rates and frequency of marketing contact.
- Keep a human review process for unusual or high-impact decisions and provide a way to correct inaccurate data.

Privacy and governance

Use data minimisation, access controls, secure storage, retention limits, and documented permissions. Follow applicable privacy and consumer-protection requirements and the organisation's policies. Do not expose individual customer records in reports or dashboards. Use aggregated reporting wherever possible. Predictions should support relevant service and marketing decisions; they should not be used to make unsupported claims about a person's health, skin condition, or wellness needs.

Model limitations

Customer behaviour can change because of seasonality, pricing, stock availability, competitor activity, product launches, economic conditions, and campaign design. Historical correlations may not remain stable. A model may also perform differently when deployed to a new channel or customer population. These limitations require ongoing monitoring and periodic revalidation.

8. Business impact and recommendations

Ifthemodelperforms reliablyand producesincrementalbusinessvalueinacontrolled pilot, it can support the following decisions in the beauty and wellness sector:

Business area	Recommended action	Measure of impact
Campaign targeting	Prioritise a limited, relevant audience using a validated score and a cost-based threshold.	Incremental conversion, cost per purchase, campaign ROI.
Product strategy	Use aggregated patterns to understand category interest and demand; validate with customer research.	Category sales, repeat purchase, product-level margin.
Customer engagement	Coordinate contact frequency and avoid repeatedly sending irrelevant promotions.	Opt-out rate, complaints, engagement, retention.
Marketing budget	Compare model-led targeting with a random or existing-rule control group.	Incremental profit and return on marketing spend.
Operations	Share aggregated demand signals with inventory and service-planning teams.	Stock availability, wastage, service utilisation.

Priority recommendations

- **Start with a pilot.** Do not deploy to the full customer base immediately. Use a limited campaign with a suitable control group.
- **Optimise for incremental value.** Compare the additional purchases and profit caused by the campaign, not just the number of customers predicted correctly.

- **Choose the threshold with stakeholders.** Finance and marketing should agree on contact costs, discount costs, expected margin and customer-experience limits.
- **Keep communication relevant.** Set frequency caps and respect consent, unsubscribe requests and channel preferences.
- **Monitor continuously.** Recheck data quality, model performance, drift and business results after launch.
- **Retain human accountability.** Marketing and product teams should review recommendations and remain responsible for decisions.

Measuring business impact

Use a randomised controlled campaign where practical. Randomly assign eligible customers to a model-targeted treatment group and a comparable control group that receives the usual approach or no campaign, as appropriate. Compare outcomes over the same period. A useful primary measure is incremental conversion rate: treatment conversion rate minus control conversion rate. Also compare incremental revenue, gross margin after discounts, contact cost, unsubscribe rate, and complaints. This design helps distinguish genuine campaign lift from purchases that would have happened anyway.

Illustrative business impact table (fill with pilot data)

Measure	Control group	Model-led group	Difference / interpretation
Eligible customers	____	____	Group sizes should be comparable.
Purchases	____	____	Count completed purchases.
Conversion rate	____%	____%	Treatment minus control = incremental lift.
Revenue / gross margin	■____	■____	Account for discounts and returns.
Contact and campaign cost	■____	■____	Include media, messaging and offer costs.
Opt-outs / complaints	____	____	Check customer-experience impact.

9. Implementation roadmap and 30–35 hour work plan

Theassignment can be completedoverapproximately33 hours. The planbelowprovides arealisticsequence from defining the business question to producing the final report. Adjust the hours to reflect the work actually completed and record evidence such as notebooks, code, model outputs, charts and review notes.

Activity	Tasks	Hours
Business framing	Define target, prediction window, stakeholders and success criteria.	3
Data understanding	Inspect schema, quality, class balance, permissions and leakage risks.	4
Baseline and modelling	Create baseline; train or review candidate model(s); document settings.	7
Validation and metrics	Cross-validation, holdout testing, metrics, threshold and error analysis.	7
Visualisation	Prepare confusion matrix, ROC/PR, calibration and performance tables.	3
Business interpretation	Translate errors into campaign costs, pilot measures and recommendations.	4
Report and review	Write, format, verify claims, add references and finalise deliverables.	5
Total		**33**

Suggested deployment stages

1. **Stage 1 – Offline evaluation:** complete data checks, baseline comparison, cross-validation and held-out testing.
2. **Stage 2 – Shadow mode:** generate predictions without using them to contact customers; compare predictions with later outcomes.
3. **Stage 3 – Controlled pilot:** use a limited campaign and control group; measure incremental outcomes and customer feedback.
4. **Stage 4 – Review and scale:** expand only if the pilot shows positive value, acceptable customer impact and operational readiness.

Monitoring plan

What to monitor	Frequency (suggested)	Action if a problem appears
Input data quality and missing values	Every scoring run / weekly review	Pause scoring if critical fields fail; investigate upstream data.
Prediction distribution and drift	Monthly	Investigate population or behaviour changes; revalidate.
Precision, recall and calibration	When outcomes mature; monthly or quarterly	Review threshold, retrain only after controlled evaluation.
Conversion, margin and customer impact	Each campaign	Stop or redesign campaigns with negative value or harm.

10. Conclusion and stakeholder summary

Predictive modelling can help beauty and wellness businessesmake moreinformed marketing and customer-engagement decisions, but its value must be demonstrated through careful evaluation. Accuracy alone does not explain whether a model is useful. Precision, recall, F1-score, AUC-ROC, PR-AUC and calibration provide complementary views of predictive performance. A staged validation plan—data audit, baseline comparison, cross-validation, final holdout testing, business pilot and ongoing monitoring—reduces the risk of relying on misleading results.

For decision-makers, the central question is not simply whether the model predicts correctly. It is whether using the predictions creates incremental profit or improves customer experience compared with the existing approach. A controlled campaign test, cost-aware threshold selection, responsible data practices and regular monitoring are therefore essential. The recommended next step is to replace the illustrative values in this report with actual model

outputs, then run a limited pilot before considering wider deployment.

Final stakeholder takeaway

Use the model asa decision-support tool, not as an unquestioned decision-maker. Validate it on unseen data, test whether it improves real campaign outcomes, protect customer privacy, and scale only when measurable business value and acceptable customer impact are demonstrated.

Appendix A: Metric formulas

Metric	Formula
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Precision	TP / (TP + FP)
Recall (Sensitivity) F1-score	TP / (TP + FN)
	2 × (Precision × Recall) / (Precision + Recall)
Specificity	TN / (TN + FP)
Incremental conversion lift	Treatment conversion rate – Control conversion rate

AUC-ROC is calculated as the area under the receiver operating characteristic curve; it is not a single threshold-based formula. PR-AUC is the area under the precision-recall curve. Report the averaging method (binary, macro, micro or weighted) when using multiclass outcomes.

Appendix B: Final submission checklist

- Replace all illustrative metric values with actual results, or clearly label the report as a proposed framework.
- Add the dataset description, target definition, sample counts, date range and class distribution.
- Include the actual confusion matrix and ROC/PR curves generated from the held-out test set.
- Document the baseline, model type, preprocessing, validation split and threshold-selection method.
- Add a feature-importance or explainability summary only if it was actually generated and validated.
- Include the controlled pilot results before claiming revenue lift or business impact.
- Check privacy, consent, fairness, spelling, page numbers and consistency.
- Submit in the file format required by the internship portal.

End of report